

Answer Key

Recommended verdict

REJECT AND ESCALATE AS SUSPECTED BUSINESS EMAIL COMPROMISE. The conclusion is based on a cluster of mutually reinforcing indicators rather than any single clue.

Evidence	Expected finding	Why it matters
From domain	rneridian-industries.example	'rn' visually imitates 'm'; classic look-alike pattern.
Reply-To	gmail.example, different from From	Replies are diverted to a separate attacker-controlled identity.
Return-Path / Message-ID	mail-relay77.example	Infrastructure does not align with the claimed corporate sender.
SPF	fail	The evaluated envelope sender was not authorised by its SPF policy in the synthetic header.
DKIM	none	The message carries no DKIM signature to verify a signing identity/integrity.
DMARC	fail	No aligned SPF or DKIM identity authenticates the visible From domain.
Received route	Earliest visible IP 198.51.100.27	The route begins outside the claimed corporate infrastructure in the scenario.
Language	Urgency, secrecy, authority, channel restriction, process bypass	These are recurring social-engineering controls in BEC.
Domain context	Three-day-old look-alike domain	Recent registration strengthens the impersonation hypothesis.
VirusTotal context	No detection is inconclusive	Threat intelligence is corroboration, not an acquittal.

Answers: SHA-256 reflection

Did one character change the entire hash? Yes. The output becomes completely different; this is the avalanche effect.

Can the hash show which character changed? No. A cryptographic hash detects difference but does not identify the changed location or reconstruct the input.

Why hash before and after analysis? To verify that the evidence or forensic copy remained unchanged during handling and examination.

What do matching values show? They provide strong evidence that the compared data is bit-for-bit identical, subject to the normal assumptions of the hash algorithm and procedure.

Tool Links and Technical Notes

[Google Admin Toolbox - Messageheader](#) - Parses SMTP message headers and helps visualise mail routing.

[CyberChef](#) - Browser-based data transformation tool; use SHA2 with Size 256 for SHA-256.

[ICANN Lookup](#) - Uses RDAP for current public domain registration data.

[VirusTotal Search](#) - Searches reports for hashes, URLs, domains, and IP addresses.

[MXToolbox SuperTool](#) - Public DNS, MX, SPF, and DMARC checks.